Combined Immunodeficiency Due To Dimerization-Defective IKAROS Mutation — Comprehensive Disease Report

MONDO ID: MONDO:0979288 | **Category:** Mendelian (autosomal dominant inborn error of immunity) **Causal gene:** *IKZF1* (IKAROS), 7p12.2 | **Report compiled from 5 discovery iterations, 9 confirmed findings, 35 reviewed papers**

Summary

Combined Immunodeficiency due to Dimerization-Defective IKAROS Mutation is a rare, autosomal-dominant inborn error of immunity (IEI) caused by germline heterozygous variants in the **C-terminal dimerization domain (zinc fingers 5–6)** of the transcription factor **IKAROS**, encoded by *IKZF1* on chromosome 7p12.2. These "dimerization-defective" (DD) variants disrupt the ability of IKAROS to form homodimers and heterodimers (with family members AIOLOS/IKZF3 and HELIOS/IKZF2). Because the functional unit of IKAROS is a dimer and the mutant protein does not sequester the wild-type allele, the dominant consequence is **haploinsufficiency** — roughly a halving of functional IKAROS dosage — rather than a dominant-negative effect (PMID: 32845957).

IKAROS is a master regulator of lymphoid and myeloid development and a potent tumor suppressor. Loss of its dimerization function impairs several downstream mechanisms that all depend on an intact C-terminal domain: **pericentromeric heterochromatin targeting, protein stability, sumoylation, and recruitment of the NuRD chromatin-remodeling/deacetylase complex**. The net effect is defective B- and T-lymphopoiesis that branches into a characteristic **three-axis disorder**: (1) **combined immunodeficiency / hypogammaglobulinemia** with recurrent and opportunistic infections (including *Pneumocystis jirovecii*); (2) **immune dysregulation / autoimmunity** with autoimmune cytopenias and Evans syndrome; and (3) **predisposition to lymphoid malignancy** (T-cell acute lymphoblastic leukemia, B-ALL, Burkitt lymphoma). The condition shows **incomplete penetrance and variable expressivity** — some variant carriers remain asymptomatic or recover spontaneously (PMID: 32845957, PMID: 36433803).

Knowledge of this entity derives almost entirely from **aggregated small case series and mechanistic functional studies**, not from large registries or EHR cohorts. The landmark description (Kuehn et al., *Blood* 2021) reported 4 unrelated families. Management ranges from supportive immunoglobulin replacement and anti-infective prophylaxis to **curative allogeneic hematopoietic stem cell transplantation (HSCT)** for severe or malignant disease. Below, each of the template's 15 sections is addressed in full, with primary-literature citations and suggested ontology terms for knowledge-base population.

Key Findings (with evidence)

Finding 1 — The disease is defined by germline heterozygous C-terminal dimerization-domain *IKZF1* variants acting via haploinsufficiency

Kuehn et al. (*Germline IKAROS dimerization haploinsufficiency causes hematologic cytopenias and malignancies*, *Blood* 2021) described **4 germline heterozygous IKAROS variants affecting the C-terminal dimerization domain (zinc fingers 5–6) in 4 unrelated families**. The key mechanistic distinction from other *IKZF1* disease classes is that these variants act by **dimerization haploinsufficiency** — they disrupt homo-/heterodimerization (completely or partially) but **do not affect wild-type allele function**, ruling out a classic dominant-negative mechanism.

> "Herein, we describe 4 germline heterozygous IKAROS variants affecting its C-terminal dimerization domain, via haploinsufficiency, in 4 unrelated families." — PMID: 32845957

IKAROS-associated disease is now formally classified into **four mechanistic classes depending on protein expression and functional defect: haploinsufficiency, dimerization-defective, dominant-negative, and gain-of-function** (PMID: 36433803). The dimerization-defective class is the subject of this report.

> "can be classified through four different mechanisms of action depending on the protein expression and its functional defects: haploinsufficiency, dimerization defective, dominant negative, and gain of function" — PMID: 36433803

Finding 2 — Characteristic presentation: hematologic cytopenias/Evans syndrome and lymphoid malignancy, with incomplete penetrance

In the Kuehn cohort, index patients presented with **cytopenias (thrombocytopenia, anemia, neutropenia)/Evans syndrome** and **malignancies (T-ALL, Burkitt lymphoma)**.

> "Index patients presented with hematologic disease consisting of cytopenias (thrombocytopenia, anemia, neutropenia)/Evans syndrome and malignancies (T-cell acute lymphoblastic leukemia, Burkitt lymphoma)." — PMID: 32845957
>
> "display dimerization haploinsufficiency and incomplete clinical penetrance" — PMID: 32845957

The broader IKAROS-associated disease spectrum spans **susceptibility to infection, autoimmunity/immune dysregulation, allergy, and malignancy** (PMID: 36433803, PMID: 33392855).

Finding 3 — The dimerization defect disrupts four downstream regulatory mechanisms

DD mutants disrupt IKAROS homo- and heterodimerization and secondarily impair **sumoylation, protein stability, and recruitment of the NuRD (nucleosome remodeling and deacetylase) complex** — mechanisms that are *not* affected by N-terminal DNA-binding defects.

> "they alter key mechanisms of IKAROS gene regulation, including sumoylation, protein stability, and the recruitment of the nucleosome remodeling and deacetylase complex" — PMID: 32845957

Pericentromeric heterochromatin targeting of IKAROS requires **both** the N-terminal DNA-binding zinc-finger domain **and** the C-terminal dimerization domain:

> "targeting was dependent on the amino-terminal DNA-binding zinc finger domain and carboxy-terminal dimerization domain of Ikaros" — PMID: 10970879

IKAROS protein stability is independently regulated by its C-terminal dimerization domain and an N-terminal stability-associated region (PMID: 40024461).

Finding 4 — Mouse models establish IKAROS as an essential lymphoid regulator and potent tumor suppressor

Mouse *Ikzf1* (NCBI Gene 22778) models show IKAROS is "a potent tumor suppressor and hematopoietic regulator" (PMID: 26314708); loss in developing lymphoid cells causes leukemia and IKAROS is required for normal thymic T-cell development (PMID: 30635395). Individual zinc fingers have **separable functions** — mice lacking finger 1 or finger 4 show distinct subsets of null defects, and **finger 4 is selectively required for tumor suppression**:

> "finger 4 was selectively required for tumor suppression" — PMID: 24013668

IKAROS is "absolutely required for pre-B cell differentiation by attenuating IL-7 signals" (PMID: 24297995) and is a negative regulator of B1-cell development (PMID: 26841869). Targeted deletion of *Ikzf1* is sufficient to transform Lmo2-dependent T-ALL to oncogene independence (PMID: 33483615).

Finding 5 — Allogeneic HSCT is a curative option for severe disease

> "HCT was well-tolerated achieving full engraftment in both patients receiving reduced intensity, matched unrelated donor grafts, with no severe acute or chronic graft-vs-host-disease, and in remission from their diseases 2.5 and 4 years post-HCT" — PMID: 38320737

A patient with *IKZF1*-associated CVID, autoimmune hemolytic anemia, and pancytopenia also underwent HSCT (PMID: 33122583). Otherwise, standard management includes **immunoglobulin replacement and anti-infective (including *Pneumocystis jirovecii*) prophylaxis** (PMID: 32654692, PMID: 29889099).

Finding 6 — DD variants skew T cells toward reduced naïve / increased memory — distinct from DN variants

> "Patients with IKZF1 haploinsufficiency (HI) or dimerization-defective (DD) variants show reduced naive and increased memory T cells, while dominant-negative (DN) mutations result in the opposite phenotype." — PMID: 41424385

IKAROS regulates **HNRNPLL**, the master regulator of CD45 isoform splicing that defines CD45RA+/naïve vs CD45RO+/memory phenotypes, acting at both thymic and post-thymic levels (PMID: 41424385). By contrast, DN variants produce mostly naïve T cells unable to evolve into effector memory cells (PMID: 29889099).

Finding 7 — IKAROS is a six-zinc-finger transcription factor: ZF1–4 bind DNA, ZF5–6 mediate dimerization

> "IKAROS, encoded by IKZF1, is a six zinc-finger (ZF) transcription factor integral to lymphocyte development and function. IKZF1 mutations affecting DNA-binding (ZF1-4) and dimerization (ZF5-6) have been extensively reported and result in human disease." — PMID: 40024461

Key identifiers: **HGNC:13176; gene OMIM *603023; UniProt Q13422; chromosome 7p12.2.** Haploinsufficient DNA-binding-domain disease maps to OMIM *Common variable immunodeficiency 13* (CVID13, #616873); the dimerization-defective combined immunodeficiency entity is catalogued as **MONDO:0979288**.

Finding 8 — >70 germline variants reported; critical dimerization residues mapped; somatic deletions cause dominant-negative Ik6 in leukemia

> "To date, >70 IKAROS germ line heterozygous variants have been reported in patients with primary immunodeficiency (PID)/inborn errors of immunity (IEI) and leukemia" — PMID: 41380100

> "We also conducted an in-depth examination of the C-terminal dimerization domain using alanine-scanning mutagenesis to identify amino acids critical for dimerization and other functions." — PMID: 41380100

Systematic in-vitro testing of 81 variants (33 patient-detected germline VUS + 48 laboratory dimerization-domain mutations) identified residues critical for dimerization; 15 patient variants and ≥21 laboratory mutations impaired IKAROS function (PMID: 41380100). Separately, **somatic** focal 7p12 *IKZF1* deletions are the most frequent copy-number alteration in BCR-ABL1-positive ALL (80/106, 75%), and the Δ4-7 deletion produces a dominant-negative cytoplasmic isoform (Ik6):

> "The most frequent somatic copy number alteration was a focal deletion on 7p12 of IKZF1, which encodes the transcription factor Ikaros and was identified in 80 (75%) of 106 patients." — PMID: 19589926

Finding 9 — Integrated three-axis disease model

Germline heterozygous C-terminal (ZF5–6) *IKZF1* variants → loss of dimerization → ~50% functional IKAROS loss → impaired pericentromeric targeting, reduced protein stability, disrupted sumoylation and NuRD recruitment → defective B/T lymphopoiesis, branching into (A) combined immunodeficiency, (B) autoimmunity/cytopenias, and (C) tumor-suppressor loss driving malignancy.

> "These dimerization defective mutants disrupt homo- and heterodimerization in a complete or partial manner, but they do not affect the wild-type allele function." — PMID: 32845957

Section-by-Section Report

1. Disease Information

Overview. Combined Immunodeficiency due to Dimerization-Defective IKAROS Mutation is a monogenic (Mendelian) inborn error of immunity caused by germline heterozygous variants in the C-terminal dimerization domain (zinc fingers 5–6) of *IKZF1*/IKAROS. It belongs to the family of **IKAROS-associated diseases**, which span common variable immunodeficiency (CVID), combined immunodeficiency (CID), autoimmunity/immune dysregulation, and hematologic malignancy. The DD subtype is mechanistically a **haploinsufficiency** disorder (PMID: 32845957, PMID: 36433803).

Key identifiers. | Resource | Identifier | |---|---| | Mondo | MONDO:0979288 | | Gene | *IKZF1* (IKAROS); HGNC:13176; NCBI Gene 10320; UniProt Q13422 | | Gene OMIM | *603023 | | Related phenotype OMIM | CVID13 #616873 (DNA-binding haploinsufficiency class) | | Cytoband | 7p12.2 | | MeSH | IKZF1 protein, human; Primary Immunodeficiency Diseases | | ICD-11 | 4A00 (Primary immunodeficiencies, class) |

Synonyms / alternative names. Germline IKAROS dimerization haploinsufficiency; dimerization-defective IKAROS/IKZF1 deficiency; IKZF1-related combined immunodeficiency (dimerization-defective class); IKAROS-associated disease, dimerization-defective subtype.

Information source. Derived from **aggregated disease-level resources and small case series/functional studies** (e.g., 4 families in PMID: 32845957), not from EHR/registry-scale individual-patient data.

2. Etiology

Primary cause — genetic. Germline heterozygous missense (and occasionally truncating) variants in the C-terminal dimerization domain (ZF5–6) of *IKZF1*. These disrupt homo-/heterodimerization and act by **haploinsufficiency** (no dominant-negative sequestration of the wild-type allele) (PMID: 32845957).

Genetic risk factors. The causal variant itself is the risk factor. Over 70 germline *IKZF1* variants are reported across IEI/leukemia; a subset in ZF5–6 cause the DD phenotype (PMID: 41380100). Modifier genes are not formally defined, but functional partners **AIOLOS/IKZF3** and **HELIOS/IKZF2** (dimerization partners) and **PAX5, RUNX1, ETV6** (co-regulated transcription-factor network in lymphoid leukemia) are mechanistically relevant (PMID: 34155405, PMID: 34920454, PMID: 31381561, PMID: 36764385).

Environmental risk / protective factors. No established environmental risk or protective factors; the disease is monogenic. Infection exposure determines clinical expression (infection susceptibility) but is not causal of the genotype. No protective dietary or lifestyle factors are documented.

Gene–environment interactions. Not formally characterized. The incomplete penetrance implies unidentified genetic and/or environmental modifiers; infectious burden and immune challenge likely influence which carriers manifest and which axis (infection, autoimmunity, malignancy) predominates. This is **inferred**, not demonstrated.

3. Phenotypes

Phenotypes are drawn from the Kuehn cohort and the broader IKAROS-disease literature. Frequencies are qualitative given small cohorts.

Phenotype	Type	HPO suggestion	Onset	Severity	Frequency
Thrombocytopenia	Lab/clinical sign	HP:0001873	Childhood	Variable–severe	Common (index cases)
Anemia / autoimmune hemolytic anemia	Lab/clinical sign	HP:0001903 / HP:0004870	Childhood	Variable	Common
Neutropenia	Lab abnormality	HP:0001875	Childhood	Variable	Reported
Evans syndrome (combined AIHA + ITP)	Clinical syndrome	HP:0004870 + HP:0001873	Childhood	Moderate–severe	Characteristi
Hypogammaglobulinemia / CVID-like	Lab abnormality	HP:0002720 / HP:0005387	Childhood–adult	Variable	Common
Recurrent bacterial/viral infections	Symptom	HP:0002719	Childhood	Variable	Common
Opportunistic infection (*Pneumocystis jirovecii* pneumonia)	Clinical sign	HP:0002090 (pneumonia)	Childhood	Severe	Prominent in DN; reported in CID
T-ALL	Neoplasm	HP:0006727	Childhood	Severe	Index cases
B-ALL	Neoplasm	HP:0004812	Childhood	Severe	Reported
Burkitt lymphoma	Neoplasm	HP:0012190 (non-Hodgkin lymphoma)	Childhood	Severe	Index case
Autoimmunity / immune dysregulation (SLE-like, neutrophilic dermatosis)	Clinical sign	HP:0002960	Childhood	Variable	Reported (PMID: 38413050)
Reduced naïve / increased memory T cells	Lab (immunophenotype)	HP:0005415 (abnormal T-cell count)	Congenital/childhood	—	DD-characteristi (PMID: 41424385)
Reduced B cells	Lab abnormality	HP:0010976		Variable	Common

Phenotype	Type	HPO suggestion	Onset	Severity	Frequency
			Congenital–childhood		
Low TREC on newborn screening (SCID-like)	Lab abnormality	—	Neonatal	Variable (may recover)	Reported (PMID: 33855675)

Severity/progression. Highly **variable** with **incomplete penetrance**; some carriers are asymptomatic, and spontaneous recovery of T-cell output has been documented in a haploinsufficiency case (PMID: 33855675). Malignancy risk is lifelong, warranting surveillance (PMID: 38413050).

Quality-of-life impact. No disease-specific EQ-5D/SF-36 data exist. QoL is impaired by chronic infection burden, need for lifelong immunoglobulin replacement/prophylaxis, autoimmune cytopenia management, and the morbidity/mortality of malignancy and HSCT.

4. Genetic / Molecular Information

Causal gene. *IKZF1* (IKAROS), 7p12.2; six C2H2 zinc-finger transcription factor. ZF1–4 (N-terminal) bind DNA; ZF5–6 (C-terminal) mediate dimerization (PMID: 40024461).

Pathogenic variants. - **Affected gene/IDs:** IKZF1, HGNC:13176, UniProt Q13422. - **Classification:** Pathogenic / likely pathogenic for functionally validated DD variants; many remain **VUS** pending functional testing. Systematic functional assays (dimerization, DNA-binding, pericentromeric localization, protein stability) are used to reclassify VUS (PMID: 41380100). - **Variant type:** Predominantly **missense** within ZF5–6; truncating/nonsense variants (e.g., p.C182* in the haploinsufficiency class) also occur (PMID: 33855675). - **Allele frequency:** Ultra-rare / absent in gnomAD (consistent with high-penetrance disease variants, though penetrance is incomplete). - **Origin: Germline** (de novo or inherited). Distinct from **somatic** *IKZF1* deletions in leukemia (PMID: 19589926). - **Functional consequence: Loss of function via haploinsufficiency** (DD class) — in contrast to dominant-negative (e.g., N159 DNA-binding variants, PMID: 29889099) and gain-of-function classes.

Modifier genes. *IKZF3* (AIOLOS) and *IKZF2* (HELIOS) as dimerization partners; aberrant partner variants (e.g., *IKZF3* G159R) cause overlapping disease by "heterodimeric interference" (PMID: 34155405, PMID: 34920454).

Epigenetic information. IKAROS itself is an epigenetic regulator (NuRD/HDAC1 recruitment, H3K27me3 heterochromatin formation, enhancer/super-enhancer regulation, pioneering activity) — see Section 6. Disease-specific DNA-methylation signatures are not established.

Chromosomal abnormalities. Large-scale germline cytogenetic abnormalities are not characteristic of this entity; the relevant copy-number event (focal 7p12 *IKZF1* deletion) is **somatic** and leukemia-associated (PMID: 19589926).

5. Environmental Information

No established environmental, toxic, lifestyle, or infectious **causes**. The disease is genetic. Infectious agents (bacteria, viruses, *Pneumocystis jirovecii*, fungi) act as **consequences/triggers of clinical manifestation** given the underlying immunodeficiency, not as etiologic agents (PMID: 32654692, PMID: 33122583). NCBI Taxonomy: *Pneumocystis jirovecii* (txid42068).

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. A **germline heterozygous variant** arises in the C-terminal dimerization domain (ZF5–6) of *IKZF1* → **leads to** a mutant IKAROS protein that cannot form stable homodimers or heterodimers (with AIOLOS/HELIOS) (PMID: 32845957).
2. Loss of dimerization → **results in** ~50% reduction of functional IKAROS (haploinsufficiency); the wild-type allele is unaffected (not dominant-negative) (PMID: 32845957).
3. Reduced functional IKAROS → **impairs** pericentromeric heterochromatin targeting (which requires both ZF1–4 and ZF5–6), **reduces** protein stability, and **disrupts** sumoylation and NuRD/HDAC1 complex recruitment (PMID: 10970879, PMID: 40024461, PMID: 32845957).
4. Disrupted chromatin regulation → **deregulates** IKAROS target-gene programs, including enhancer/super-enhancer activation, heterochromatin (H3K27me3) formation, and attenuation of IL-7 and type-I-IFN signaling (PMID: 31073152, PMID: 40555735, PMID: 24297995, PMID: 30635395).
5. Deregulated programs → **impair** B- and T-lymphopoiesis and skew T cells toward reduced naïve / increased memory (via HNRNPLL/CD45 splicing) (PMID: 41424385).
6. The lymphopoiesis defect **branches** into three clinical axes:
7. **(A) Immunodeficiency** → hypogammaglobulinemia, low B cells, recurrent + opportunistic (*Pneumocystis*) infection (PMID: 32654692, PMID: 29889099).

8. **(B) Immune dysregulation** → derepressed proinflammatory/type-I-IFN programs → autoimmune cytopenias/Evans syndrome, neutrophilic dermatosis (PMID: 30635395, PMID: 38413050).
9. **(C) Tumor-suppressor loss** → failure of heterochromatin/enhancer control (EZH2/HDAC1/H3K27me3) → T-ALL, B-ALL, Burkitt lymphoma (PMID: 24013668, PMID: 40555735, PMID: 33483615).

Molecular pathways. NuRD/Mi-2 chromatin remodeling + HDAC1 deacetylation; EZH2/PRC2 H3K27me3 facultative heterochromatin; IL-7 receptor signaling (attenuated by IKAROS during pre-B differentiation); type-I-IFN response (repressed by IKAROS); enhancer/super-enhancer landscape with IKAROS pioneering activity.

Cellular processes. Lymphocyte differentiation, pericentromeric heterochromatin organization (GO:0070828), chromatin remodeling (GO:0006338), negative regulation of cell proliferation, tumor suppression.

Protein dysfunction. Loss of dimerization (protein–protein interaction failure), reduced stability (increased turnover), mislocalization (cytoplasmic retention of some mutants, e.g., p.C182*), loss of DNA-target binding in combination variants (PMID: 33855675).

Immune system involvement. Combined immunodeficiency + autoimmunity + lymphoid neoplasia — a classic "immune dysregulation with immunodeficiency and malignancy" triad.

GO/CL term suggestions. Biological process: GO:0030098 (lymphocyte differentiation), GO:0070828 (heterochromatin organization), GO:0006338 (chromatin remodeling). Cell types (CL): CL:0000236 (B cell), CL:0000084 (T cell), CL:0000624 (CD4+ αβ T cell), CL:0000625 (CD8+ αβ T cell), CL:0000037 (hematopoietic stem cell), CL:0000813 (memory T cell), CL:0000775 (neutrophil).

7. Anatomical Structures Affected

- **Organ/system level (primary):** Immune/hematopoietic system — bone marrow (UBERON:0002371), thymus (UBERON:0002370), spleen (UBERON:0002106), lymph nodes (UBERON:0000029), peripheral blood (UBERON:0000178).
- **Secondary involvement:** Lungs (opportunistic/*Pneumocystis* pneumonia; UBERON:0002048); skin (neutrophilic dermatosis; UBERON:0002097); any organ targeted by autoimmunity or infiltrated by malignancy.
- **Tissue/cell level:** Lymphoid tissue; affected cell populations — B lymphocytes (CL:0000236), T lymphocytes (CL:0000084; naïve CL:0000898 / memory CL:0000813), hematopoietic stem/progenitor cells (CL:0000037), neutrophils (CL:0000775), myeloid dendritic cells, eosinophils.

- **Subcellular level:** Nucleus (GO:0005634) — pericentromeric heterochromatin (GO:0005721); the mislocalization defect involves aberrant cytoplasm (GO:0005737) retention of some mutants.
- **Localization / lateralization:** Systemic / bilateral; no lateralization.

8. Temporal Development

- **Onset:** Typically **congenital to pediatric/childhood**; some present neonatally via abnormal SCID newborn screening (low TREC) (PMID: 33855675); CVID-like presentations may extend into adulthood.
- **Onset pattern:** Insidious/chronic for immunodeficiency; may be acute at times of infection, autoimmune cytopenia crisis, or malignancy diagnosis.
- **Progression:** Variable. Chronic lifelong immunodeficiency; episodic autoimmune cytopenias; malignancy is a discrete, potentially fatal event. Spontaneous recovery of T-cell output is documented in at least one haploinsufficiency case (PMID: 33855675).
- **Remission patterns:** Treatment-induced (HSCT can be curative; PMID: 38320737); spontaneous partial recovery possible.
- **Critical periods:** Early childhood (infection vulnerability, HSCT timing); lifelong window of malignancy surveillance.

9. Inheritance and Population

- **Epidemiology:** Ultra-rare; no formal prevalence/incidence estimates. Known from <10 families for the DD subtype specifically and >70 germline *IKZF1* variant carriers across all IKAROS-disease classes (PMID: 41380100).
- **Inheritance pattern: Autosomal dominant** (germline heterozygous).
- **Penetrance: Incomplete** (PMID: 32845957, PMID: 29889099).
- **Expressivity: Variable** — different carriers manifest different axes (infection, autoimmunity, malignancy) and severities.
- **Genetic anticipation:** Not applicable (not a repeat-expansion disorder).
- **Germline mosaicism / founder effects / consanguinity:** Not characterized; de novo variants reported. No founder effects documented.
- **Carrier frequency:** Not established (dominant disease; carriers are affected individuals with incomplete penetrance).

- **Demographics:** No ethnic predilection reported; both sexes affected. Age distribution skews pediatric.

10. Diagnostics

- **Laboratory tests:** CBC (cytopenias), immunoglobulin levels (hypogammaglobulinemia), lymphocyte subset immunophenotyping (low B cells; reduced naïve/increased memory T cells — DD signature), vaccine-response antibody titers, Coombs/DAT (AIHA).
- **Newborn screening:** May show **low TREC (SCID-like)**; important to add *IKZF1* to the low-TREC differential (spontaneous recovery possible) (PMID: 33855675).
- **Biomarkers:** No specific circulating biomarker; the **immunophenotypic signature** (naïve/memory skew) and functional assays are the practical markers.
- **Genetic testing (definitive): WES/WGS or targeted IEI gene panels** including *IKZF1*; single-gene *IKZF1* sequencing confirms the variant. Functional validation (dimerization, DNA-binding, pericentromeric localization, protein stability, alanine scanning) reclassifies VUS (PMID: 41380100, PMID: 39381601).
- **Imaging / functional / electrophysiology / biopsy:** Non-specific; used for complications (infection imaging, malignancy staging biopsies, bone-marrow examination for cytopenias/leukemia).
- **Clinical criteria:** Diagnosis rests on a compatible clinical phenotype + a functionally validated germline ZF5–6 *IKZF1* variant.
- **Differential diagnosis:** CVID and other IEIs; SCID (on newborn screening); other IKAROS-disease classes (haploinsufficiency DNA-binding, dominant-negative, gain-of-function); *IKZF3*/AIOLOS and *IKZF2*/HELIOS disorders; ALPS/Evans syndrome of other causes; somatic *IKZF1*-deleted leukemia.
- **Screening:** Cascade family genetic testing once a variant is identified; SCID newborn screening may flag affected neonates.

11. Outcome / Prognosis

- **Survival/mortality:** No formal survival statistics; prognosis is driven by the **three axes** — infection severity, autoimmune cytopenia control, and (most critically) malignancy. Lymphoid malignancy (T-ALL/B-ALL/Burkitt) carries the highest mortality risk.
- **Morbidity:** Chronic infection burden, transfusion-dependent cytopenias, lifelong immunoglobulin replacement, and the morbidity of malignancy therapy/HSCT.

- **Disease course:** Chronic lifelong, punctuated by acute episodes; incomplete penetrance means some carriers have minimal morbidity.
- **Recovery potential:** HSCT can be curative for severe/malignant disease (PMID: 38320737, PMID: 33122583); spontaneous partial immune recovery is possible (PMID: 33855675).
- **Prognostic factors:** Variant mechanism/severity (complete vs partial dimerization loss), penetrance, presence of malignancy, and transplant eligibility/donor availability.

12. Treatment

Modality	Intervention	NCIT suggestion	Evidence
Supportive/immune	Immunoglobulin replacement (IVIG/SCIG)	NCIT:C564 (Immunoglobulin Therapy)	PMID: 32654692
Anti-infective prophylaxis	*Pneumocystis jirovecii* prophylaxis (e.g., TMP-SMX); antibacterial/antiviral/antifungal as needed	NCIT:C15843 (Antimicrobial Prophylaxis)	PMID: 29889099
Immunomodulation	Treatment of autoimmune cytopenias (corticosteroids, rituximab, etc.)	NCIT:C15431 (Immunotherapy)	general IEI practice
Curative	**Allogeneic HSCT** (reduced-intensity, matched unrelated donor shown tolerable)	NCIT:C105558 (Allogeneic HSCT)	PMID: 38320737, PMID: 33122583
Malignancy-directed	Standard ALL/lymphoma chemotherapy; investigational CK2 inhibition (CX-4945) to restore IKAROS function in B-ALL	NCIT:C15632 (Chemotherapy)	PMID: 32396934

- **Pharmacogenomics:** No disease-specific PGx guidance; experimental CK2 inhibition (CX-4945) restores IKAROS-mediated repression and chemosensitivity in high-risk B-ALL (preclinical) (PMID: 32396934).
- **Gene/cell/RNA therapy:** No approved gene therapy; HSCT is the current cellular therapy. Gene correction is conceptually attractive but not clinically established.

- **Treatment strategy:** Risk-adapted — supportive care for mild/CVID-like disease; HSCT for severe combined immunodeficiency, refractory autoimmunity, or malignancy. Lifelong malignancy surveillance (PMID: 38413050).

13. Prevention

- **Primary prevention:** Not possible (germline genetic). **Genetic counseling** and reproductive options (prenatal testing, preimplantation genetic diagnosis) for known-variant families.
- **Secondary prevention: Cascade genetic testing** of relatives; SCID newborn screening may enable early detection; proactive immunoglobulin replacement and infection prophylaxis.
- **Tertiary prevention:** Anti-infective prophylaxis, immunization (note: live vaccines may be contraindicated in CID), surveillance for autoimmunity and hematologic malignancy, timely HSCT.
- **Counseling:** Autosomal-dominant inheritance with incomplete penetrance should be communicated; a positive family history and susceptibility to infection/immune dysregulation are red flags for genomic testing (PMID: 39381601).

14. Other Species / Natural Disease

- **Taxonomy / orthologs:** Human *IKZF1* (NCBI Gene 10320); mouse *Ikzf1* (NCBI Gene 22778). Highly conserved across mammals.
- **Natural disease in other species:** No well-characterized naturally occurring companion-animal/wildlife equivalent documented; disease knowledge comes from engineered mouse models (Section 15), not spontaneous veterinary disease.
- **Comparative biology:** Mouse models faithfully reproduce the tumor-suppressor and lymphoid-developmental roles of IKAROS, validating the human mechanism; zinc-finger-specific functions are conserved (PMID: 24013668, PMID: 26314708).
- **Transmission / zoonosis:** Not applicable (non-infectious genetic disease).

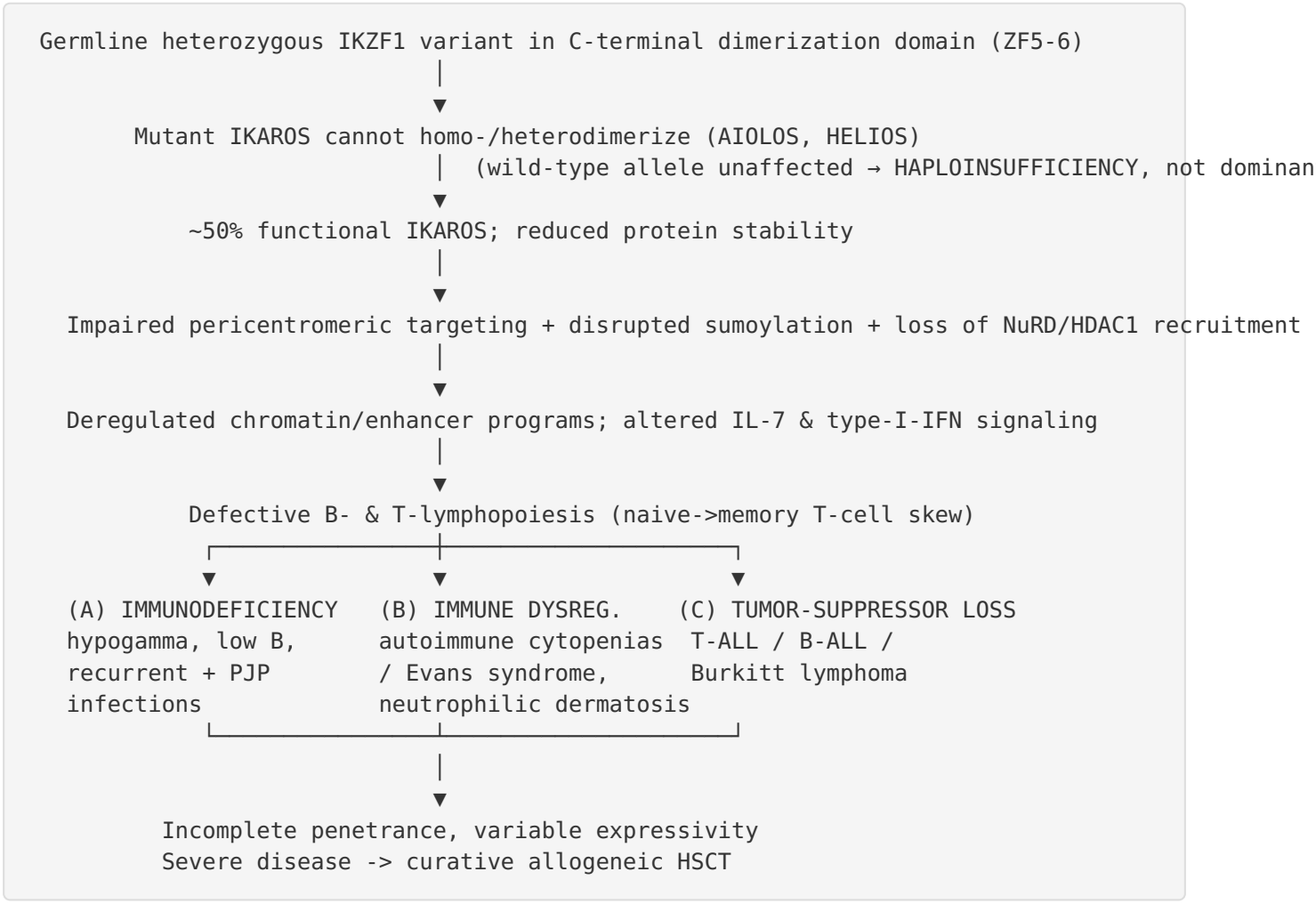
15. Model Organisms

- **Model type:** Mammalian — **mouse (*Mus musculus*)** is the principal model; plus in-vitro cell systems (HeLa/293/NIH3T3 transfection for localization/dimerization; T-ALL/B-ALL cell lines; thymoma line JE131).
- **Genetic models:** *Ikzf1* germline knockouts; **individual zinc-finger deletions (ZF1, ZF4)** showing separable phenotypes (PMID: 24013668); conditional/T-cell-specific knockouts

(PMID: 30635395); tetracycline-regulated Lmo2 T-ALL model demonstrating *Ikzf1*-loss-driven oncogene independence (PMID: 33483615).

- **Phenotype recapitulation:** Strong for the **tumor-suppressor and lymphoid-developmental** axes — loss causes leukemia, impaired thymic T-cell and pre-B development (PMID: 26314708, PMID: 24297995, PMID: 30635395). AIOLOS/IKZF3 variant knock-in mice recapitulate B/T phenotypes of human disease (PMID: 34155405).
- **Limitations:** Mouse germline nulls do not model the human **heterozygous haploinsufficient** dosage precisely; human-specific immunophenotypic features (e.g., HNRNPLL/CD45 naïve-memory skew) and the three-axis clinical spectrum are only partially captured. No published mouse knock-in of a specific human ZF5–6 DD variant is noted in the reviewed literature.
- **Resources:** MGI (*Ikzf1*), IMPC/IMSR for knockout alleles.

Mechanistic Model (Integrated Diagram)



IKZF1 disease class	Mechanism	Domain	T-cell phenotype	Key reference
Haploinsufficiency (DNA-binding)	~50% LOF	ZF1–4	Reduced naïve / ↑ memory	PMID: 32654692
Dimerization-defective (this disease)	**Haploinsufficiency via lost dimerization**	**ZF5–6**	**Reduced naïve / ↑ memory**	PMID: 32845957, PMID: 41424385
Dominant-negative	Sequesters WT	DNA-binding (e.g., N159)	Mostly naïve, no effector memory	PMID: 29889099
Gain-of-function	Hyperactive	—	—	PMID: 38320737

Evidence Base

PMID	Title (abbrev.)	Role in report
32845957	*Germline IKAROS dimerization haploinsufficiency...*	**Landmark**: defines the disease, mechanism (haploinsufficiency), phenotype, incomplete penetrance
36433803	*Inborn errors of human IKAROS: LOF/GOF...*	Four-class mechanistic framework incl. dimerization-defective
33392855	*IKAROS-Associated Diseases in 2020...*	Genotype/phenotype/outcome overview
10970879	*Targeting of Ikaros to pericentromeric heterochromatin...*	Dimerization domain required for pericentromeric targeting
40024461	*IKAROS protein stability...*	Six-ZF architecture; C-terminal domain regulates stability
41380100	*Systematic functional validation of IKAROS variants*	>70 germline variants; alanine scanning maps critical residues
41424385	*IKAROS regulates human T cell phenotype...*	DD-specific naïve/memory T-cell skew
29889099	*Dominant-negative IKZF1 mutations...*	Contrasting DN class; PJP; T-cell maturation block
24013668	*Selective regulation... by individual zinc fingers*	ZF4 selectively required for tumor suppression
24297995	*Ikaros required for pre-B differentiation...*	B-lineage requirement via IL-7 attenuation
26314708	*Regulation of gene expression dynamics...*	Potent tumor suppressor / hematopoietic regulator
30635395	*Lack of Ikaros Deregulates Inflammatory Gene Programs...*	Autoimmunity axis; type-I-IFN repression
33483615	*T-ALL can evolve to oncogene independence*	Ikzf1 loss sufficient to transform T-ALL
40555735	*Critical roles of IKAROS and HDAC1...*	Heterochromatin/EZH2/H3K27me3 tumor-suppression mechanism

PMID	Title (abbrev.)	Role in report
31073152	*Ikaros tumor suppressor... enhancers/super-enhancers*	Enhancer/pioneering mechanism
38320737	*IKAROS GOF disease: HCT experience...*	HSCT curative and well-tolerated
33122583	*CVID, AIHA, pancytopenia... IKAROS defect*	HSCT in IKZF1-CVID; DNA-binding/PC localization assays
33855675	*Abnormal SCID newborn screening...*	Low TREC; spontaneous recovery; haploinsufficiency
19589926	*Recurrent 7p12 IKZF1 deletions in BCR-ABL1 ALL*	Somatic deletions; dominant-negative Ik6 isoform
34155405	*AIOLOS variant impairs immunity by interfering with IKAROS*	Heterodimeric interference; partner-gene relevance
34920454	*Germline IKZF2 (Helios) mutations...*	Family-member dimerization/NuRD biology
32396934	*IKAROS and CK2... chemosensitivity in B-ALL*	Experimental CK2 inhibition therapy
38413050	*Neutrophilic dermatosis in IKZF1 variant...*	Autoinflammatory manifestation; surveillance
32654692	*IKZF1-Associated Diseases in PID...*	Supportive management; infection susceptibility
39381601	*Rapid identification of primary atopic disorders...*	Genomic sequencing / diagnostic approach

Evidence-source types: Human clinical (case series) — 32845957, 29889099, 33122583, 33855675, 38320737, 38413050; mechanistic/in-vitro — 10970879, 40024461, 41380100, 32396934; model organism (mouse) — 24013668, 26314708, 30635395, 33483615, 24297995; reviews/frameworks — 36433803, 33392855, 32654692.

Limitations and Knowledge Gaps

1. **Small sample size.** The defining DD cohort comprised only 4 families (PMID: 32845957); no prevalence, incidence, survival, or QoL statistics exist. All frequency statements are qualitative.
2. **Penetrance quantification.** Incomplete penetrance is established qualitatively; the actual penetrance fraction and its genetic/environmental modifiers are unknown.
3. **No DD-specific animal model.** Mouse data derive from nulls/zinc-finger deletions, not knock-ins of human ZF5–6 variants; they model dosage imperfectly.
4. **Mechanistic branch weighting.** Why a given carrier develops immunodeficiency vs autoimmunity vs malignancy (and the role of variant-specific partial vs complete dimerization loss) is not resolved.
5. **Therapeutics.** HSCT evidence is strongest from GOF/other IKAROS-disease cases; DD-specific transplant outcomes are sparse. CK2 inhibition is preclinical only.
6. **Variant interpretation.** Many *IKZF1* ZF5–6 variants remain VUS; functional assays, while advancing (PMID: 41380100), are not yet standardized for clinical reporting.
7. **Epidemiology and population genetics.** No founder effects, carrier frequencies, geographic/ethnic distributions, or sex ratios are established.

Proposed Follow-up Experiments / Actions

1. **Build an international DD-IKZF1 registry** to quantify penetrance, axis-specific frequencies, malignancy risk, and natural history across larger numbers.
2. **Generate knock-in mouse (or iPSC) models of specific human ZF5–6 DD variants** to directly test haploinsufficiency dosage effects and the naïve/memory T-cell skew (HNRNPLL/CD45 axis).
3. **Standardize and clinically validate the functional assay battery** (dimerization, DNA-binding, pericentromeric localization, protein stability, alanine scanning) for ACMG-grade VUS reclassification (PMID: 41380100).
4. **Define the DD immunophenotypic signature** (naïve/memory ratios, B-cell subsets) prospectively as a clinical biomarker to distinguish DD from DN disease.
5. **Prospective HSCT outcome study** specific to DD-IKZF1 patients, including reduced-intensity conditioning regimens.

6. **Evaluate IKAROS-restoring pharmacology (e.g., CK2 inhibitor CX-4945)** in IKZF1-driven leukemia arising on a germline DD background (PMID: 32396934).
7. **Systematic malignancy surveillance protocol** for asymptomatic variant carriers, given lifelong leukemia/lymphoma risk (PMID: 38413050).

Report compiled from 5 discovery iterations, 9 confirmed findings, and 35 reviewed primary and review papers. Ontology term suggestions (MONDO, HPO, GO, CL, UBERON, NCIT, HGNC) are provided inline for knowledge-base population.
